



SAVE TIME. STAY PREPARED.
WE'VE GOT YOUR BACK!



EMERGENCY SUB PLANS

$$E=mc^2$$

MIDDLE SCHOOL SCIENCE

VOLUME 3

10
NEW
LESSONS

Ten more no-prep emergency lessons for the day you cannot be there.

No lab. No equipment.



NO-PREP
Just print
& go



READY-TO-USE
Complete plans
& materials



FOR BUSY DAYS
Engaging,
easy to follow

- ✓ 10 brand new lessons, none repeated from Volumes 1 or 2
- ✓ No lab, no equipment, no science background needed
- ✓ Every lesson has the words to say and a timing table
- ✓ Runs entirely on paper and a pencil
- ✓ Full answer keys that explain why, not just what

INSIDE THIS VOLUME

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**QUALITY
RESOURCES.
CONFIDENT
CLASSROOMS.**

*Created by
a teacher,
for teachers.*



How This Pack Works

START HERE

Ten complete emergency science lessons. Each runs for a full class period, needs nothing but photocopies, and is written so a substitute with no science background can teach it confidently.

This is Volume 3. All ten lessons are new. Nothing here repeats Volume 1 or Volume 2, so the volumes sit side by side in the drawer and never give a class the same lesson twice.

What every lesson contains

PAGE	WHAT IT IS	WHO IT IS FOR
1	Teacher briefing. Timing table, the exact words to say, the three questions students will ask with the answers written out, and what to do with early finishers.	The substitute. Keep this page.
2	Reading or task page. The material for the lesson, with numbered paragraphs and any diagram needed.	Copy for students.
3	Question page. Six to nine questions with writing space built in.	Copy for students.
4	Go further page. Extension task and a challenge for students who finish early.	Copy for students.
5	Answer key. Every answer explained, plus marking notes and the wrong answers to expect.	The substitute. Keep this page.

ABSOLUTELY NO LAB, AND THAT IS THE POINT

Most schools will not let a substitute run a practical, and rightly so. Equipment and safety supervision need the regular teacher present. That is exactly why science sub plans are hard to find.

Nothing in this pack is heated, poured, cut or plugged in. Every lesson runs on paper. The quick start page says so plainly, and each opening script pre-empts the inevitable question about doing an experiment with a reason that is actually true.

Reading level and grouping

Passages sit in the grades 6 to 8 band. Every lesson works as independent silent work, as partner work, or read aloud by the substitute. The briefing page tells the substitute how to run all three, so no decision is left to guesswork on the day.

HOW THIS PACK RELATES TO THE EARLIER VOLUMES

Volume 1 covers fair tests and variables, cell structure, forces and motion, the periodic table, energy transfers and efficiency, food webs, the particle model and changes of state, the water cycle and weather, plate tectonics, and evaluating data and claims. **Volume 2** covers physical and chemical change, wave properties, heredity and Punnett squares, natural selection, photosynthesis and respiration, the circulatory and respiratory systems, density and buoyancy, series and parallel circuits, moon phases and seasons, and rock layers and relative dating.

No topic is repeated across the series.

Substitute Quick Start

5 MINUTES • FOR THE SUBSTITUTE

If you have five minutes before the bell, read this page and nothing else. It is enough.

In the first five minutes

- 1 Find the lesson the teacher left, or pick any lesson from the index. They are independent. Order does not matter.
- 2 Copy the three student pages, one set per student.
- 3 Keep the briefing page and the answer key. **Do not photocopy those.**
- 4 Read the briefing page. It has a timing table and the words to say.

DO NOT OPEN THE CUPBOARDS

Nothing in this pack needs equipment, and you should not take anything out of a science storeroom, fridge or cupboard. If a student tells you the class always does a practical on this day, the honest answer is that safety supervision needs their regular teacher present.

How the period runs, every time

MINUTES	WHAT HAPPENS
0-5	Students settle. You read the opening lines from the briefing page out loud. Hand out the student pages.
5-15	Students read the task page. Silent, partner, or read aloud by you. Your choice.
15-40	Students work through the question page. You circulate.
40-50	Anyone finished moves to the Go Further page. Collect everything with two minutes to go.

IF A STUDENT ASKS YOU SOMETHING YOU CANNOT ANSWER

Check the briefing page first. The three most common questions for that lesson are already answered there.

If it is not there, this always works and is always true: **"Write down your best answer and put a star next to it. Your teacher will look at the starred ones."**

Lesson Index

ALL TEN LESSONS IN THIS VOLUME

Lessons are numbered 21 to 30 so they sit alongside the earlier volumes without clashing. Every lesson is independent. Teach them in any order.

#	LESSON	FOCUS	NGSS	TIME
21	Bending the Beam	Light: reflection, refraction and how we see	MS-PS4-2	40-50 min
22	Fields You Cannot See	Magnets, magnetic fields and electromagnets	MS-PS2-3	45-55 min
23	The Scale That Sorts	Acids, bases, the pH scale and neutralization	MS-PS1-2	40-50 min
24	Made, Not Found	Synthetic materials, polymers and trade-offs	MS-PS1-3	40-50 min
25	The Gap Before You Move	Nervous system, stimulus and response, reflexes	MS-LS1-8	45-55 min
26	The Long Journey	Digestive system, enzymes and absorption	MS-LS1-3	45-55 min
27	Very Small Enemies	Microbes, disease, immunity and vaccines	MS-LS1-1	45-55 min
28	The Same Bones	Evidence for common ancestry	MS-LS4-3	45-55 min
29	Why Nothing Falls Down	Gravity, orbits and the solar system	MS-ESS1-2	45-55 min
30	How a Mountain Becomes Sand	Weathering, erosion and deposition	MS-ESS2-2	40-50 min

Which lesson should I leave?

- **Planned absence, want real work done:** Lesson 23 or 29. Both produce calculation you can grade.
- **Emergency, class you do not know well:** Lesson 21 or 30. Both are self-contained with a low frustration ceiling.
- **Restless group, last period:** Lesson 22 or 25. Both have a strong hook and a visible finish line.
- **Out several days:** 26 then 27 run well as a pair, and 28 then 29 do the same for the big-picture lessons.

NGSS codes are given as the closest matching middle school performance expectation. This resource is independently created and is not affiliated with or endorsed by any state agency or standards body.

Bending the Beam

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students trace rays through a mirror and a glass block, then use the results to explain a straw that looks broken and a pool that looks shallow. The lesson also corrects the belief that seeing is something the eye does to an object, which is far more common at this age than teachers expect.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the two ray diagrams.
13-24	Questions 1 to 4. The angle table reads straight off the rule.
24-38	Questions 5 to 8, which need explaining rather than measuring.
38-48	Go Further: the pool that is deeper than it looks.
48-50	Collect all pages.

SAY THIS TO START

Light goes in straight lines. That sounds too obvious to be worth saying, and it is the single most useful fact in the whole topic. Only two things in the universe can change the direction of a beam of light: a surface that bounces it, or a change in what it is traveling through. Everything today is one of those two, including why a straw looks broken and why a swimming pool is always deeper than it looks.

The three questions students will ask

THEY ASK	YOU SAY
"Why does a straw look bent in water?"	Light slows down when it enters water, so it changes direction. Your brain still assumes it came in a straight line, so it puts the straw in the wrong place.
"Where is the image in a mirror, really?"	Nowhere. No light ever goes behind the mirror. The light only <i>appears</i> to come from that point, which is why it is called a virtual image.
"Do your eyes send out a beam?"	No. Light travels from a source, bounces off the object, and goes into your eye. If eyes sent out beams you could read in a completely dark room.

FINISHED EARLY?

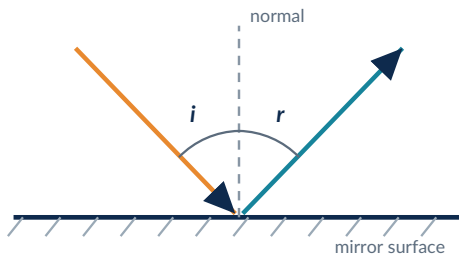
The Go Further page gives four real and apparent pool depths. Students divide one by the other, find the same number every time, and then use it to predict a fifth.

Read: Two Ways Light Changes Direction

BEFORE YOU READ

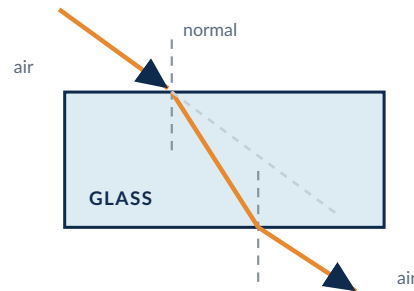
Study the two diagrams below carefully. Questions 2 and 6 on the next page depend on being able to read them, so it is worth a full minute before you start.

REFLECTION



Angle in (i) always equals angle out (r).

REFRACTION



Slower in glass, so it bends toward the normal.

1 Light travels in straight lines. It keeps going in a straight line until something makes it change direction, and only two things can do that. A surface can bounce it, which is called **reflection**. Or it can cross into a different material and change speed, which is called **refraction**. Every single thing in this lesson is one of those two.

2 Reflection follows one rule, and it has no exceptions. The ray arrives at a surface, and it leaves at exactly the same angle it arrived at. Both angles are measured from a line drawn at right angles to the surface, called the **normal**, and not from the surface itself. Arrive at 30 degrees to the normal and you leave at 30 degrees to the normal.

3 So why does a mirror show your face when a sheet of white paper does not? Both reflect light. The difference is smoothness. A mirror is smooth, so rays that arrive in an orderly pattern leave in an orderly pattern and the pattern survives as an image. Paper is rough at a scale far too small to see, so every ray meets a surface tilted a slightly different way and they scatter in all directions. The light is still reflected. The order is not.

4 That scattering is also the reason you can see the paper at all. You see this page because light from a lamp or a window hits it, scatters, and some of it goes into your eye. Objects that make their own light are rare: the Sun, a flame, a screen, a bulb. Everything else you have ever looked at was borrowing light from something else.

5 It is worth being blunt about the alternative, because a lot of people quietly believe it. Eyes do not send out beams. If they did, a completely dark room would be no problem. The reason you cannot see in the dark is that there is no light arriving to be bounced.

6 Now refraction. Light does not travel at the same speed in everything. It is fastest in empty space, slightly slower in air, noticeably slower in water and slower still in glass. When a ray crosses the boundary at an angle, one side of it slows before the other, and the ray swings round. Going into a slower material it bends *toward* the normal. Coming back out into a faster one it bends away again.

7 Your brain has no idea any of this happened. It receives light and assumes, as it always does, that it arrived in a perfectly straight line. So it traces the ray back along the wrong path and puts the object where that path came from. The straw is not bent and the pool bottom has not risen. Both are exactly where they always were, and you are looking in slightly the wrong place.

8 Bend light on purpose and it becomes useful. A lens is a curved piece of glass shaped so that every ray passing through it is refracted by just the right amount to meet the others at one point. That is the whole idea behind glasses, cameras, microscopes, telescopes, and the lens sitting in the front of your own eye right now.

Questions: Bending the Beam

Answer in the space provided. Questions 2 and 6 refer to the diagrams on the reading page.

1 There are only two things that can change the direction of a ray of light. Name both and give an everyday example of each.

2 Complete the table using the reflection rule.

ANGLE OF INCIDENCE (FROM THE NORMAL)	ANGLE OF REFLECTION
20°	
35°	
52°	
64°	
A ray arriving straight along the normal	

3 A mirror and a sheet of white paper both reflect light, but only one shows your face. Explain the difference.

4 Explain why you can see this page even though it does not make any light of its own.

5 A student says their eyes send out a beam that lets them see. Give one everyday observation that proves this cannot be true.

6 Light going from air into glass bends toward the normal. What does that tell you about the speed of light in glass compared with air?

7 Explain, using refraction, why a straw standing in a glass of water looks broken at the water line.

Go Further: The Pool That Is Deeper Than It Looks

EXTEND

THE PROBLEM

Every year people drown in water they thought was shallower than it was. A pool always looks less deep than it is, and the effect gets worse the deeper the water. Below are four measurements taken at different points along the same pool.

POINT	REAL DEPTH	DEPTH IT APPEARS TO BE	REAL ÷ APPARENT
A	4.0 ft	3.0 ft	
B	6.0 ft	4.5 ft	
C	8.0 ft	6.0 ft	
D	10.0 ft	7.5 ft	

- 1 Complete the last column. Write down what you notice about the four answers.
- 2 The deep end is 12 feet deep. Use your answer to question 1 to predict how deep it will look. Show your working.
- 3 Explain, using refraction, why the bottom of the pool appears closer to the surface than it really is.

CHALLENGE

People who fish with a spear learn very quickly that if you aim straight at the fish you can see, you miss every time. Should they aim above the fish or below it? Explain your answer using what you now know about how light leaves the water. Two or three sentences.

IF YOU STILL HAVE TIME

A periscope uses two flat mirrors, each tilted at 45 degrees, to let you see over a wall. Sketch the path of one ray from the object at the top to the eye at the bottom, and mark the two points where the reflection rule is doing the work.

Answer Key: Bending the Beam

QUESTION PAGE

1. Reflection, where a surface bounces the ray, and refraction, where the ray changes speed crossing into a different material. Any sensible examples: a mirror, a puddle, a shiny spoon; a straw in water, a lens, a pool bottom.
2. 20° , 35° , 52° , 64° . A ray arriving along the normal has an angle of incidence of 0° , so it reflects at 0° and comes straight back the way it came.
The last row separates students who learned the rule from students who copied the pattern. Accept "straight back" as a full answer.
3. A mirror is smooth, so rays leave in the same orderly arrangement they arrived in and the image survives. Paper is rough at a very small scale, so rays scatter in all directions and no image is left.
Most students write that paper is "not shiny". Push for the word rough, and for the idea that the paper reflects just as much light — it just disorders it.
4. Light from a lamp or a window hits the page, scatters off it in all directions, and some of that scattered light enters the eye.
5. The obvious one: you cannot see in a completely dark room. If eyes produced the light, darkness would make no difference.
Also accept: you cannot see round corners; closing the curtains changes what you can see; a torch helps and it would not need to.
6. Light travels slower in glass than in air. Bending toward the normal is what happens when a ray slows down.
7. Light from the underwater part of the straw is refracted as it leaves the water and enters the air. The brain assumes the light traveled in a straight line, so it traces it back to the wrong place and the underwater part appears shifted. The straw itself is perfectly straight.
Full credit needs both halves: the light bends, and the brain assumes it did not.

GO FURTHER PAGE

POINT	REAL	APPARENT	REAL ÷ APPARENT
A	4.0	3.0	1.33
B	6.0	4.5	1.33
C	8.0	6.0	1.33
D	10.0	7.5	1.33

1. Every answer is the same, about 1.33. Water always makes a depth look about three quarters of what it really is.
 2. $12 \div 1.33 =$ about 9 ft. The deep end looks three feet shallower than it is.
Accept 9 or 9.0. A student who multiplies instead of divides gets 16 ft — ask them whether water is supposed to make things look deeper or shallower.
 3. Light leaving the water bends away from the normal as it speeds up in air. The eye traces that light back in a straight line, and that straight line meets the bottom higher up than the real bottom, so the pool looks shallower.
- Challenge.** Aim below the fish. The fish appears higher in the water than it really is, for exactly the same reason the pool bottom does, so the real fish is deeper than the image you are aiming at.

This question sorts out who understood the pool and who memorised it. Any answer that says "below, because the real one is deeper than it looks" is full credit.

MARKING SHORTCUT

If you are short on time, mark question 3 and Go Further question 3. Between them they show whether the student can explain both halves of the topic in their own words. Question 2 and the table are self-checking — a wrong number is obvious at a glance.

Fields You Cannot See

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

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Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students sort materials into magnetic and not magnetic, work out what happens when a magnet is cut in half, and then read two sets of electromagnet data to find what makes one stronger. The lesson exists mainly to separate two words that students treat as identical: metal and magnetic.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the page and study the field diagram.
14-26	Questions 1 to 3. The sorting table is the main work.
26-40	Questions 4 to 7.
40-53	Go Further: the two electromagnet data tables.
53-55	Collect all pages.

SAY THIS TO START

I am going to say something that sounds wrong. Almost nothing is magnetic. Out of everything in this room, a magnet will stick to a very short list, and most of the metal in here is not on it. Aluminum cans, copper wire, gold, the coins in your pocket — a magnet ignores all of them. So today, forget the word metal. It is the wrong word and it has been misleading you.

The three questions students will ask

THEY ASK	YOU SAY
"Are all metals magnetic?"	No, and this is the big one. Only iron, nickel, cobalt and steel, which is mostly iron. Aluminum, copper, gold and silver are all metals and none of them are magnetic.
"What if you cut a magnet in half?"	You get two complete magnets, each with a north and a south pole. Cut those and you get four. You can never end up with a single pole on its own.
"Is an electromagnet a real magnet?"	Yes, while the current is flowing. It has poles and a field just like a bar magnet. Switch the current off and the magnetism goes with it, which is the whole point of it.

FINISHED EARLY?

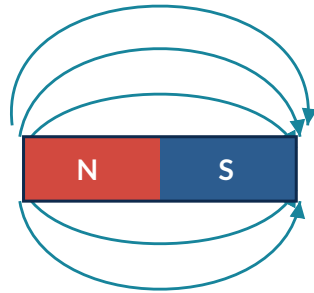
The Go Further page has two data tables from a scrapyard crane. Students find the pattern in each, then predict a value that is not in the table.

Read: The Force That Works Across a Gap

BEFORE YOU READ

The diagram below shows two things: the shape of the invisible field around a single magnet, and what happens when two magnets are brought close. Question 3 depends on the right-hand half.

THE FIELD AROUND A BAR MAGNET



Field lines always run out of N and back into S.

TWO MAGNETS, TWO RESULTS



S faces N — unlike poles pull together



N faces N — like poles push apart

- 1 A magnet pulls on certain things without touching them. The region around it where that pull can be felt is called a **magnetic field**. You cannot see it, but it has a definite shape, and iron filings scattered on paper over a magnet will line up and draw it for you.
- 2 Every magnet has two ends that behave differently, called the **north** and **south poles**. Bring the north of one magnet near the south of another and they pull together. Bring two norths together and they push apart, hard. Unlike poles attract, like poles repel. That one sentence answers most questions about magnets.
- 3 Now the part that surprises people. Almost nothing is magnetic. The list of everyday materials a magnet will stick to is essentially iron, nickel, cobalt, and steel, which is mostly iron with a little carbon added. Aluminum is a metal and is not magnetic. Copper is a metal and is not magnetic. Neither are gold, silver, brass or lead. Treating *metal* and *magnetic* as the same word will get you the wrong answer more often than the right one.
- 4 Here is a strange one. Cut a bar magnet in half and you do not get a north piece and a south piece. You get two smaller magnets, each with its own north and its own south. Cut those and you get four. Keep going and it never stops working. Nobody has ever isolated a single magnetic pole, and it is not for want of trying.
- 5 The Earth itself behaves like an enormous magnet with a field stretching far out into space. A compass needle is nothing more than a small magnet free to swing, so it lines itself up with that field, which is why it always settles pointing the same way. Sailors were navigating with this a thousand years before anyone could explain it.
- 6 Magnetism and electricity turn out to be two sides of one thing. Send a current through a wire and a magnetic field appears around it, and it vanishes the instant the current stops. Wind the wire into a coil and the fields of all the loops add together into something much stronger. That is an **electromagnet**.
- 7 There are three ways to make one stronger, and they can all be tested on a bench. Increase the current. Add more turns of wire to the coil. Or put a piece of iron down the middle as a core, which concentrates the field dramatically and is by far the biggest single improvement of the three.
- 8 But strength is not really the point. A permanent magnet is always on, and something that is always on cannot let go. An electromagnet can be switched off, which is why the crane in a scrapyard picks up a car and drops it exactly where it wants, and why the same idea runs doorbells, electric motors, loudspeakers, hard drives and the machine that takes an MRI scan.

Questions: Fields You Cannot See

Answer in the space provided.

1 In your own words, what is a magnetic field?

2 Sort each object. Write Y if a magnet will stick to it and N if it will not, then give your reason.

#	OBJECT	Y OR N	REASON
1	Steel paperclip		
2	Aluminum drink can		
3	Copper wire		
4	Iron nail		
5	Plastic ruler		
6	Gold ring		
7	Steel scissor blade		
8	Glass marble		

3 Using the right-hand diagram, describe what happens in each case and say why.

WHAT YOU DO	WHAT HAPPENS	WHY
North brought near south		
North brought near north		

4 A bar magnet is cut cleanly in half through the middle. Describe the two pieces you end up with.

5 Name the three ways of making an electromagnet stronger.

6 Give one job an electromagnet can do that a permanent magnet could never do, and explain why.

Go Further: The Crane in the Scrapyard

THE SITUATION

A scrapyard crane lifts cars with an electromagnet. Before building the full-size version, the engineers tested a small one in a workshop and counted how many steel paperclips it could hold. They changed one thing at a time and kept everything else the same.

Test 1. Current held steady at 1.0 A. Number of turns of wire changed.

TURNS OF WIRE	20	40	60	80	100
Paperclips held	6	13	19	25	31

Test 2. Turns held steady at 60. Current changed.

CURRENT	0.5 A	1.0 A	1.5 A	2.0 A
Paperclips held	9	19	28	38

- Describe the pattern in Test 1 in one sentence. Use numbers from the table.

- Predict how many paperclips 120 turns would hold. Explain how you worked it out.

- In both tests the engineers changed one thing and kept everything else the same. Why does that matter? What would go wrong if they changed the turns and the current together?

- The workshop magnet had no iron core. The full-size crane has a huge one. What effect would you expect that to have, and why?

CHALLENGE

A permanent magnet needs no electricity at all, never fails in a power cut, and costs nothing to run. A scrapyard could easily afford a very strong one. Explain why every scrapyard in the world uses an electromagnet instead. Two or three sentences.

IF YOU STILL HAVE TIME

A compass is left sitting next to the crane's electromagnet while the power is off, and it points north as usual. Predict what the needle does the moment the crane is switched on, and what it does when the crane is switched off again.

Answer Key: Fields You Cannot See

QUESTION PAGE

1. The region around a magnet where its pull or push can be felt. It cannot be seen but it has a definite shape, running out of the north pole and back into the south.
2. Y: paperclip, iron nail, steel scissor blade (all steel or iron). N: aluminum can, copper wire, gold ring, plastic ruler, glass marble.
Mark the reason column, not the letter. Three of the five "no" answers are metals, and that is the point of the question. A student who writes "no, because it is not a metal" for the aluminum can has got the right letter for the wrong reason and has missed the lesson.
3. North near south: they pull together, because unlike poles attract. North near north: they push apart, because like poles repel.
4. Two smaller complete magnets, each with its own north pole and its own south pole. You do not get a north piece and a south piece.
The common wrong answer is exactly that — one north half and one south half. It is a very reasonable guess and worth saying so before correcting it.
5. More current, more turns of wire in the coil, and an iron core down the middle.
6. It can be switched off, so it can put down whatever it picked up. Any sensible example: a scrapyard crane, a door release, a sorting line that drops steel into one bin.

60 FURTHER PAGE

1. More turns means more paperclips, and it rises steadily — about 6 more clips for every extra 20 turns.
2. About 37. The pattern adds roughly 6 clips per 20 turns, so 100 turns held 31 and 120 turns should hold about $31 + 6 = 37$.
Accept anything from 36 to 38 with reasoning shown. A student who doubles 31 because 120 is close to double 60 has spotted a real pattern in the wrong place — worth half credit and a conversation.
3. Because if two things change at once you cannot tell which one caused the result. Changing turns and current together would give a bigger magnet but no way of knowing which change did the work.
4. It would get much stronger. An iron core concentrates the magnetic field, and of the three improvements it makes the biggest difference.

Challenge. Because a permanent magnet cannot let go. A crane has to release the car over the right pile, and a magnet that is always on would hold onto it. Being able to switch it off is the entire reason the job works.

Full credit for any answer built on "it needs to be able to drop things". Strong students add that a power cut would drop the load, which is a genuine safety problem engineers design around.

IF A STUDENT FINISHES EVERYTHING

Ask them this: iron filings sprinkled around a magnet form curved lines rather than straight ones. What does the shape of those lines tell you that a simple "it attracts things" does not?

The Scale That Sorts

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students place substances on the pH scale, then work with the fact that the scale is not evenly spaced: each step of one is a factor of ten. The lesson deliberately uses food for most of its acids, because the belief that acid means dangerous is what stops students reasoning about pH properly.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the pH scale.
13-26	Questions 1 to 3. The sorting table reads off the scale.
26-38	Questions 4 to 6. Question 4 is the factor-of-ten calculation.
38-48	Go Further: the garden bed.
48-50	Collect all pages.

SAY THIS TO START

You have already eaten several acids today. Orange juice is an acid. Vinegar is an acid. Cola is a stronger acid than either. And there is acid in your stomach right now, strong enough to dissolve metal, sitting quite happily inside you. So whatever the word acid means, it does not mean dangerous. Today you will find out what it actually measures.

The three questions students will ask

THEY ASK	YOU SAY
"Are all acids dangerous?"	No. Lemon juice is a stronger acid than most things in a school lab and you can drink it. What matters is how strong it is and how concentrated, not the word acid.
"Are alkalis the safe ones then?"	No, and that surprises people. Oven cleaner and drain cleaner are alkalis and they will burn you badly. Both ends of the scale are the dangerous ends.
"Why isn't pH 6 half as acidic as pH 3?"	Because each step of one is ten times, not one unit. pH 3 is a thousand times more acidic than pH 6. This is on the student page and it is the hardest idea in the lesson.

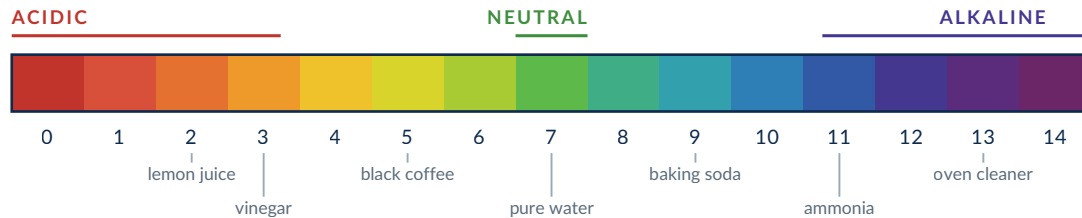
FINISHED EARLY?

The Go Further page gives a garden bed measured at pH 5.0 and a table of what different plants need. Students work out what will grow now, and what will grow after lime is added.

Read: A Number Line for Chemicals

BEFORE YOU READ

The scale below is the whole lesson in one picture. Questions 1 and 2 read straight off it, so study it before you start.



1 Chemists sort a great many substances into two families. **Acids** are one, **bases** are the other, and a base that dissolves in water is usually called an **alkali**. The **pH scale** puts a number on where something sits, running from 0 at the sour end to 14 at the slippery end.

2 Below 7 is acidic. Exactly 7 is neutral, and pure water sits there. Above 7 is alkaline. That is the entire rule, and everything else in this lesson is a consequence of it.

3 You find out where something sits using an **indicator**, which is a substance that changes color depending on what it is put into. Universal indicator gives a different color for every number on the scale, which is where the colors in the diagram come from. Red cabbage does the same job well enough that it is a standard kitchen demonstration.

4 Now look at where the food is. Lemon juice is pH 2. Vinegar is 3. Cola sits at about 2.5, which makes it more acidic than vinegar. Coffee is 5. Your own stomach runs somewhere between 1.5 and 3.5, and it is doing that on purpose. Acids are not a laboratory thing that got loose. They are most of what you ate this week.

5 The other end deserves the same honesty in reverse. Baking soda is a mild alkali at pH 9 and you can bake with it. Oven cleaner is around 13 and will destroy skin. Drain cleaner is worse. Strong alkalis are every bit as dangerous as strong acids, and arguably worse, because they do not hurt immediately and people keep their hands in them for longer.

6 Here is the part that catches everybody. The scale is not evenly spaced. Each step of one unit is a factor of **ten**. So pH 4 is not slightly more acidic than pH 5, it is ten times more acidic. pH 3 is a hundred times more acidic than pH 5, and a thousand times more than pH 6. A change of two on this scale is an enormous change in the real world, which is exactly why scientists get alarmed by numbers that look small.

7 Put an acid and a base together and each cancels the other out, moving the mixture toward 7. That is **neutralisation**, and it is behind more everyday things than you would guess. An indigestion tablet is a base swallowed to neutralise stomach acid that has gone where it should not. Toothpaste is mildly alkaline to deal with the acid bacteria produce after you eat sugar. Farmers spread lime on fields that have turned too acidic for the crop.

8 Living things are fussy about pH in a way that seems out of proportion until you remember the factor of ten. Human blood is held between 7.35 and 7.45, and drifting a few tenths outside that range is a medical emergency. Rivers, soil and swimming pools are all monitored for the same reason: a number that moves a little has moved a lot.

Questions: The Scale That Sorts

Questions 1 and 2 refer to the pH scale on the reading page.

1 Complete the table. Write A for acidic, N for neutral or K for alkaline.

SUBSTANCE	PH	A, N OR K	SUBSTANCE	PH	A, N OR K
Lemon juice	2		Baking soda	9	
Pure water	7		Vinegar	3	
Oven cleaner	13		Milk	6.5	
Black coffee	5		Seawater	8.1	

2 Put these five in order, from most acidic to most alkaline: milk, oven cleaner, lemon juice, pure water, black coffee.

3 A student writes: “Acids are the dangerous ones and alkalis are safe.” Give two separate reasons this is wrong.

4 How many times more acidic is pH 2 than pH 5? Show how you worked it out.

5 Explain what happens chemically when someone with indigestion takes an antacid tablet.

6 After a sugary drink, bacteria in your mouth produce acid. Explain why a dentist cares about that, and how a mildly alkaline toothpaste helps.

Go Further: The Garden Bed

THE SITUATION

A gardener tests the soil in a new bed and gets **pH 5.0**. Different plants need different soil, and a plant outside its range will survive but grow badly.

PLANT	PH RANGE IT NEEDS
Blueberry	4.5 – 5.5
Potato	5.0 – 6.0
Carrot	6.0 – 6.8
Cabbage	6.5 – 7.5
Lavender	7.0 – 8.0

- The soil is at pH 5.0 today. Which of the five plants would grow well, and which would not?

- The gardener digs in lime, which is a base. The soil is retested at **pH 6.6**. Which plants would grow well now?

- One plant was fine before the lime and is now outside its range. Which one, and what should the gardener do about it?

- Explain, using the word neutralisation, what the lime actually did to the soil.

CHALLENGE

Clean rainwater is naturally about pH 5.6, not 7, because carbon dioxide from the air dissolves into it. A lake affected by pollution is measured at pH 4.6. That is only one unit lower, and yet fish begin to die. Explain why a difference of one on this scale is far bigger than it looks. Two or three sentences.

IF YOU STILL HAVE TIME

The gardener's neighbor tips a large amount of lime on in one go and the soil reaches pH 8.2. Which plants can they now grow, and why is over-correcting just as much of a problem as never correcting at all?

Answer Key: The Scale That Sorts

QUESTION PAGE

1. Acidic: lemon juice, vinegar, black coffee, milk. Neutral: pure water. Alkaline: baking soda, oven cleaner, seawater.

Milk at 6.5 and seawater at 8.1 are the two that catch people, because both are close to 7 and neither feels like an acid or an alkali. The rule is the number, not the feeling.

2. Lemon juice (2), black coffee (5), milk (6.5), pure water (7), oven cleaner (13).

3. First, plenty of acids are harmless — lemon juice, vinegar and cola are all acids and all edible. Second, plenty of alkalis are dangerous — oven cleaner and drain cleaner will burn skin badly. Danger depends on how strong and how concentrated something is, not which side of 7 it sits on.

Two distinct reasons are needed. "Some acids are safe" twice is half credit.

4. A thousand times. Each step is ten times, and there are three steps from 5 down to 2, so $10 \times 10 \times 10 = 1000$.

The common wrong answer is 3, or sometimes 30. If several students give it, write $10 \times 10 \times 10$ on the board — it usually lands immediately.

5. The tablet is a base. It neutralises the excess stomach acid, moving the pH up toward 7 and stopping the burning feeling.

6. The acid the bacteria produce attacks tooth enamel, and enamel does not grow back. A mildly alkaline toothpaste neutralises that acid, bringing the mouth back toward neutral before damage is done.

60 FURTHER PAGE

1. Blueberry (4.5–5.5) and potato (5.0–6.0) are both fine at 5.0. Carrot, cabbage and lavender are all outside their range and would grow badly.

2. At 6.6, carrot (6.0–6.8) and cabbage (6.5–7.5) are now fine. Potato is just outside at the top. Lavender still needs 7.0 or above.

3. Blueberry. It needed 4.5–5.5 and the soil is now 6.6, well outside. The gardener should move it, or grow it in a separate bed or pot with untreated acidic soil.

This is the best question on the page. It requires holding the whole table in mind at once rather than checking one plant. Accept any practical answer that separates the blueberry from the limed soil.

4. Lime is a base, so it neutralised some of the acid in the soil, cancelling it out and moving the pH up toward 7.

Challenge. Because each step of one on the pH scale is a factor of ten. Going from 5.6 to 4.6 means the water is ten times more acidic, not a small amount more. That is a large enough change to damage fish eggs and gills.

This is the conceptual heart of the lesson and worth marking carefully. Full credit requires the factor of ten, not just "it is worse than it looks".

MARKING SHORTCUT

Mark question 4 and the Challenge only. Both test the factor-of-ten idea, which is the one thing from this lesson that students get wrong for years afterwards. Everything else on the pages is practice toward it.

Made, Not Found

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students sort materials into natural and synthetic, learn what a polymer actually is, then choose materials for two very different jobs using a property table. The lesson ends on the honest version of the plastic problem: the property that makes it useful is the same property that makes it a problem.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the chain diagram.
13-26	Questions 1 to 3. The sorting table is quick.
26-38	Questions 4 to 6, which need reasoning.
38-48	Go Further: choosing a material for two jobs.
48-50	Collect all pages.

SAY THIS TO START

Look at what you are wearing. Some of it grew — cotton came off a plant, wool came off a sheep. And some of it never existed anywhere in nature until a chemist decided it should. Polyester, nylon, the sole of your shoe. Today you are going to find out what those made-up materials actually are, why they were worth inventing, and why the exact thing that makes them brilliant is also the thing that makes them a problem.

The three questions students will ask

THEY ASK	YOU SAY
"What is a polymer?"	A very long chain made of one small unit repeated thousands of times. Poly means many, mer means part. It is on the student page with a diagram.
"Is plastic made from oil?"	Yes, almost all of it. Crude oil is separated into different parts, and some of those parts are the small units that get joined into plastic chains.
"Isn't biodegradable plastic the answer?"	Not on its own. Most of it only breaks down in an industrial composter at high temperature. In a park bin or the ocean it behaves much like ordinary plastic.

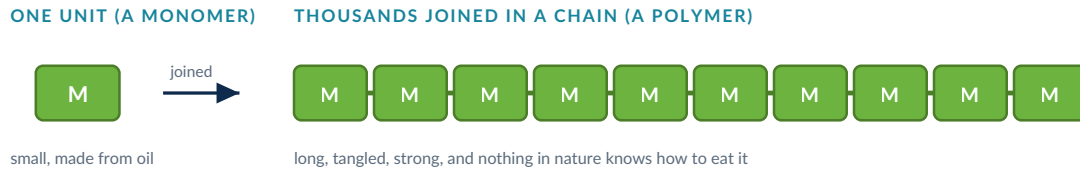
FINISHED EARLY?

The Go Further page gives a table of four materials and two jobs with very different requirements. Students choose and justify, which produces genuinely markable reasoning.

Read: Materials Nobody Found

BEFORE YOU READ

One word explains most of this lesson. The diagram below shows what it means, and question 2 asks you to put it in your own words.



1 For most of human history we used what we could find. Wood, stone, clay, wool, cotton, leather, bone. Each of those comes with a fixed set of properties, and if the properties did not quite suit the job, you put up with it. Wood rots. Wool shrinks. Leather cracks. There was no menu.

2 A **synthetic** material is one that does not occur in nature at all. Chemists take a raw material, almost always crude oil, break it into small pieces and join those pieces back together into an arrangement that has never existed before. The result can be given properties chosen in advance, which is the genuinely new thing about it.

3 Most of these materials are **polymers**. The word is worth taking apart: *poly* means many, *mer* means part. A polymer is one small unit repeated over and over into a chain thousands of units long. Not all polymers are invented. Cellulose in wood is a natural polymer, and so is the DNA inside every cell you own.

4 How the chains are made and packed decides everything about how the material behaves. Long loose chains that slide over each other give you a stretchy shopping bag. The same basic chemistry with the chains packed tightly and lined up gives you a rigid drink bottle. Same family, completely different object, and the difference is arrangement rather than ingredients.

5 This is why materials are now chosen backwards, starting from the job. A bike helmet needs to be light, and it needs to crush slowly rather than shatter. A surgical glove needs to be thin enough to feel through, strong enough not to tear, and cheap enough to throw away after one patient. A climbing rope needs to stretch slightly under load rather than snap. In each case somebody wrote down the properties first and then built a material to match.

6 Now the part that gets skipped. Plastics last because their chains are strong and unfamiliar. No bacterium or fungus has ever encountered a polythene chain in its evolutionary history, so nothing has evolved the tools to break one down. That is precisely why your drink bottle does not dissolve in the rain. It is also precisely why it is still recognisably a bottle in four hundred years.

7 So the useful property and the harmful property are not two separate things you could fix independently. They are the same property. Any honest discussion of plastic has to start there rather than treating durability as an unfortunate side effect.

8 Which is also why *biodegradable* on a label deserves a second look. Most biodegradable plastics need an industrial composting facility running at 130 to 140 degrees Fahrenheit before they break down at any useful speed. Dropped in a park bin, buried in landfill or washed into the sea, a great many of them behave much like ordinary plastic. The word is doing more work in the marketing than in the material.

Questions: Made, Not Found

Answer in the space provided.

1 Sort each material. Write N for natural or S for synthetic.

MATERIAL	N OR S	MATERIAL	N OR S	MATERIAL	N OR S
Cotton		Nylon		Wool	
Polyester		Leather		Polystyrene	
Silk		Rubber from a tree		Acrylic	

2 Explain what a polymer is. Use the word *units* in your answer.

3 A shopping bag and a drink bottle can be made from very similar chemistry, yet one is floppy and one is rigid. Explain how that is possible.

4 For each job, name the property that matters most and suggest why a natural material was replaced.

THE JOB	PROPERTY THAT MATTERS MOST	WHY NOT A NATURAL MATERIAL?
Bike helmet		
Surgical glove		
Climbing rope		

5 Explain why the property that makes plastic so useful is the same property that makes it a problem in the environment.

6 A label says “biodegradable”. Give one reason that might not mean what a shopper assumes it means.

Go Further: Choosing the Right Material

THE TASK

You are choosing a material for two very different products. Use the table, then justify your choice — the reasoning matters more than the answer.

MATERIAL	WEIGHT	STRENGTH	COST PER POUND	WATERPROOF	YEARS TO BREAK DOWN
Polypropylene	Light	High	\$0.95	Yes	20 – 30
Stainless steel	Heavy	Very high	\$2.10	Yes	50 – 200
Glass	Heavy	Low (shatters)	\$0.63	Yes	1 million +
Coated paperboard	Very light	Low	\$0.50	Briefly	Under 1

1 **Job A.** A reusable water bottle a student will carry every day for four years. Which material, and why? Name two properties from the table that decided it.

2 **Job B.** A sterile wrapper for a single hospital syringe. It is opened once and thrown away immediately. Which material, and why?

3 The two jobs led to different answers even though both need to be waterproof. Explain what made the difference.

CHALLENGE

One bottle is advertised as **made from 100% recycled plastic**. Another is advertised as **100% recyclable**. They sound almost identical. Explain what each one actually promises, and say which claim you think is worth more. Two or three sentences.

IF YOU STILL HAVE TIME

Glass takes about a million years to break down — far longer than any plastic in the table — and yet nobody talks about a glass crisis in the oceans. Suggest two reasons why.

Answer Key: Made, Not Found

QUESTION PAGE

1. Natural: cotton, wool, leather, silk, rubber from a tree. Synthetic: nylon, polyester, polystyrene, acrylic.

Rubber is the one to watch. Natural rubber comes from a tree; synthetic rubber is made from oil. The question specifies the tree, so it is natural — but a student who queries it has noticed something real.

2. A polymer is a very long chain made of one small unit repeated thousands of times. Poly means many, mer means part.

3. Because the properties depend on how the chains are arranged rather than what they are made of. Long loose chains that slide past each other give something floppy. Chains packed tightly and lined up give something rigid.

4. Helmet: light and crushes slowly instead of shattering — no natural material combines both. Glove: thin, strong and cheap enough to discard after one use. Rope: stretches slightly under load rather than snapping, and does not rot when wet.

Accept any well-argued property. The reasoning column is where the marks are.

5. Plastic is useful because nothing breaks it down, so it stays strong and waterproof for years. Nothing breaks it down because no organism has evolved the tools to attack those chains. That is the same fact stated twice — you cannot remove one without removing the other.

Full credit needs the recognition that these are one property, not two. “It is strong and it also pollutes” is half credit.

6. Most biodegradable plastics only break down in an industrial composter at high temperature. In landfill, a park bin or the ocean they behave much like ordinary plastic.

60 FURTHER PAGE

1. Stainless steel or polypropylene, both defensible. Steel: very high strength and long life, and weight barely matters for a bottle carried in a bag. Polypropylene: light and strong at half the cost. Reject glass (shatters) and paperboard (only briefly waterproof).

There is no single right answer here. Mark the two properties named and whether they actually support the choice.

2. Coated paperboard. It is used once and thrown away, so strength and lifespan do not matter, and it is the cheapest and breaks down in under a year. Choosing steel for a wrapper thrown away after ten seconds is the error the question is testing.

3. How long the product has to last. Job A needs a material that survives four years of daily use. Job B needs one that survives about ten seconds, so the sensible choice is the one that disappears fastest afterwards.

This is the question worth marking. It is the whole point of the page: the right material depends on the job, not on which material is “best”.

Challenge. “Made from recycled plastic” means something has already happened — old plastic was actually used to make this bottle. “Recyclable” only means it *could* be recycled if the right facility exists and somebody puts it in the right bin, which often does not happen. The first claim is the stronger one.

Any answer that spots that one describes something already done and the other describes a possibility gets full credit.

The Gap Before You Move

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. Keep: this page and the answer key.

What this lesson does

Students order the six stages of a reflex arc, identify receptors and effectors for three everyday situations, and then work with real reaction-time data. The lesson is built around one fact students find genuinely startling and can verify from memory: your hand leaves the hot pan *before* you feel it.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the page and study the pathway diagram.
14-28	Questions 1 to 3. The ordering task is the core.
28-40	Questions 4 to 6.
40-53	Go Further: the reaction time data.
53-55	Collect all pages.

SAY THIS TO START

Think about the last time you touched something hot. Your hand was already moving before it hurt. That is not a figure of speech and it is not you misremembering — it is literally the order the events happened in, and there is a good reason for it. Today you are going to work out why your body makes that decision without asking your brain first.

The three questions students will ask

THEY ASK	YOU SAY
“Does the message go to the brain?”	Not first. In a reflex the decision is made in the spinal cord. A message is sent up to the brain at the same time, but it arrives after your hand has already moved.
“Why not just use the brain for everything?”	Distance and steps. Going via the brain is a longer trip with more processing, and the delay is the difference between a red mark and a serious burn.
“How fast do nerve signals actually go?”	The fastest are around 250 feet per second, which is quick but nowhere near instant. Fast enough that the delays in this lesson are measured in fractions of a second.

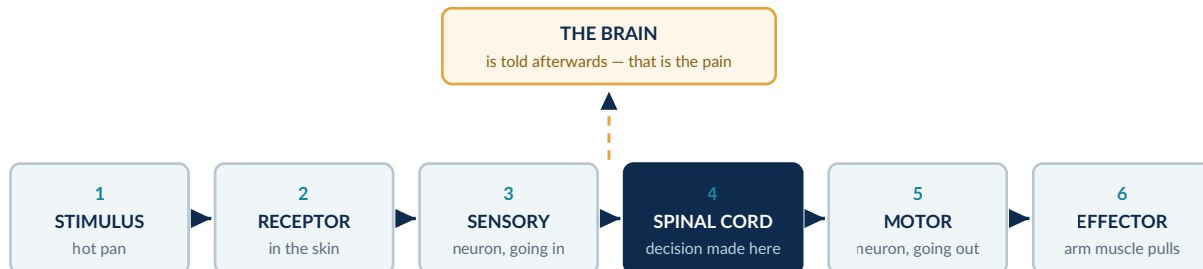
FINISHED EARLY?

The Go Further page has ruler-drop data from three students, tested focused and then distracted. Students convert distances to times using a supplied table and calculate the average difference.

Read: The Decision Your Brain Did Not Make

BEFORE YOU READ

The diagram below is the route a reflex takes. Question 1 asks you to put the same six stages in order without looking, so study it now.



The hand is already moving by the time the message reaches the brain. The response does not wait for it.

1 Everything your body does in response to the world runs through the same six-stage route. Something happens (the **stimulus**). A specialized cell detects it (the **receptor**). A **sensory neuron** carries the signal inward. A decision is made. A **motor neuron** carries the instruction back out. And a muscle or gland (the **effector**) does something about it (the **response**).

2 Neurons are cells built for one job: carrying an electrical signal along their length, quickly. The fastest travel at around 250 feet per second. That is fast, but it is not instant, and every extra inch of route costs time.

3 Receptors are specialized for the thing they detect and are useless for anything else. The retina in your eye detects light and cannot detect sound. The skin has separate receptors for pressure, for temperature and for damage. Your tongue and nose detect chemicals. Nothing detects everything.

4 Now the interesting part. For most responses the decision genuinely is made by the brain. You see a ball coming, your brain works out where it will be, and you put your hand there. That is flexible and clever and comparatively slow.

5 A **reflex** skips the brain entirely. The sensory neuron carries the signal to the spinal cord, and the spinal cord connects it straight to a motor neuron going back out. No thinking is involved and no permission is asked. Your hand is pulled off the pan by a decision made somewhere in your back.

6 A message is also sent up to the brain at the same moment, but it has further to travel and more processing to get through. It arrives a fraction of a second later, and that arrival is the moment you feel the pain. So the sequence really is: touch, move, then hurt. You have experienced this many times and probably never noticed the order.

7 The saving is roughly a tenth of a second, which sounds trivial and is not. On a hot surface, a tenth of a second is the difference between a red mark and a burn that needs a dressing. Blinking when something comes at your eye, coughing, and your pupils shrinking in bright light are all the same design: too important to wait for a decision.

8 For everything else, the slower route is worth it. A reflex is fast because it is fixed, and fixed responses are useless in a situation nobody has met before. Thinking costs you a fraction of a second and buys you the ability to do something you have never done.

Questions: The Gap Before You Move

Answer in the space provided.

1 Put these six stages in the correct order, numbering them 1 to 6. Follow the signal from a hot pan to a hand pulling away.

ORDER	STAGE
	A motor neuron carries the instruction outward
	Heat receptors in the skin detect the temperature
	The hand pulls away from the pan
	A sensory neuron carries the signal to the spinal cord
	The arm muscle contracts
	The fingers touch the hot pan

2 Complete the table for three everyday situations.

SITUATION	RECEPTOR	EFFECTOR	RESPONSE
Bright light hits your eye			
A ball flies at your face			
You touch something sharp			

3 Explain why your hand moves away from a hot pan *before* you feel the pain.

4 Why does a reflex not go via the brain? Give the reason in terms of time.

5 Catching a ball is not a reflex, even though it feels fast. Explain how you can tell the difference.

6 Reflexes are faster than thinking. Suggest why your body does not simply use reflexes for everything.

Go Further: Measuring the Gap

THE SITUATION

A ruler is held above someone's open hand and dropped without warning. They catch it as fast as they can, and the distance it fell before being caught is measured. A longer fall means a slower reaction. Each student was tested twice: once concentrating, and once while holding a conversation.

Use this table to turn distances into times.

DISTANCE FALLEN	2 IN	4 IN	6 IN	8 IN	10 IN	12 IN
Reaction time	0.10 s	0.14 s	0.18 s	0.20 s	0.23 s	0.25 s

1 Complete the results table.

STUDENT	FOCUSED	TIME	DISTRACTED	TIME	EXTRA TIME
A	6 in		10 in		
B	8 in		12 in		
C	4 in		8 in		

2 Work out the average extra time caused by being distracted. Show your working.

3 A car traveling at 60 mph covers 88 feet every second. Using your average from question 2, how much further does that car travel before the driver reacts?

4 Every student was slower when distracted, but by slightly different amounts. Does that weaken the conclusion? Explain.

CHALLENGE

Pulling your hand off a hot pan takes about **0.02 seconds**. Catching a falling ruler takes about **0.20 seconds** — ten times longer. Both involve detecting something and moving a muscle. Explain what makes one ten times slower than the other. Two or three sentences.

Answer Key: The Gap Before You Move

QUESTION PAGE

- 1 The fingers touch the hot pan. 2 Heat receptors in the skin detect the temperature. 3 A sensory neuron carries the signal to the spinal cord. 4 A motor neuron carries the instruction outward. 5 The arm muscle contracts. 6 The hand pulls away from the pan.
- Bright light: retina in the eye / muscles of the iris / pupil gets smaller. Ball at your face: retina / eyelid muscles / you blink. Something sharp: pain receptors in the skin / arm muscle / hand pulls away.
Accept any correct receptor-effector pairing. The common error is naming the eye as the effector for the first one — the eye detects, the iris muscle acts.
- Because the reflex decision is made in the spinal cord, which is a much shorter trip than going to the brain. The message to the brain is sent at the same time but has further to travel and more processing to pass through, so it arrives after the hand has already moved. Feeling the pain is that message arriving.
This is the question that carries the lesson. Full credit needs both the short route and the fact that the brain is informed later rather than not at all.
- Going via the brain means a longer route and extra processing, which costs roughly a tenth of a second. On a hot surface that tenth of a second is the difference between a red mark and a real burn.
- Catching a ball needs a judgement — where the ball will be, which hand to use, when to close your fingers. That judgement can only be made in the brain, so it is a voluntary response. A reflex is always the same fixed action and you cannot choose not to do it.
- Because a reflex is fast only because it is fixed, and a fixed response is no use in a situation you have never met. Thinking is slower but it can produce an action that has never been done before.

60 FURTHER PAGE

STUDENT	FOCUSED	DISTRACTED	EXTRA
A	6 in = 0.18 s	10 in = 0.23 s	0.05 s
B	8 in = 0.20 s	12 in = 0.25 s	0.05 s
C	4 in = 0.14 s	8 in = 0.20 s	0.06 s

- $(0.05 + 0.05 + 0.06) \div 3 = 0.16 \div 3 =$ about **0.05 seconds**.
- $88 \times 0.05 =$ about **4.4 feet**. The car travels more than four extra feet before the driver even begins to react.
Worth saying out loud: four and a half feet is roughly the length of the car's hood. Students find that number much more convincing than the decimal.
- No. Every single student got slower, which is the pattern that matters. Exact amounts differ because people differ, but the direction of the change was the same in all three cases.
Strong students point out that three people is a small sample and more results would be better. That is excellent scientific thinking — give extra credit.

Challenge. The hot pan is a reflex, so the signal only travels to the spinal cord and straight back out. Catching the ruler is voluntary: the signal has to reach the brain, be recognized, be turned into a decision, and then travel back down. It is a much longer route with far more processing in the middle.

This is the payoff of the whole lesson. Any answer contrasting the short spinal route with the long brain route gets full credit, whether or not the word voluntary appears.

IF A STUDENT FINISHES EVERYTHING

Ask them why a professional sprinter is not allowed to be credited with a start faster than 0.10 seconds after the gun. What are the officials assuming about the human body, and are they right to assume it?

The Long Journey

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students follow food through a thirty-foot tube and work out where it actually enters the body, which is not where most of them assume. The lesson is built on one sentence that reliably stops a class dead: the sandwich in your stomach is still, technically, outside you.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the page.
14-28	Questions 1 to 3. The organ table is the main task.
28-40	Questions 4 to 6.
40-53	Go Further: the shortened intestine data.
53-55	Collect all pages.

SAY THIS TO START

Here is something that sounds like a trick and is not. The sandwich you ate at lunch is not inside your body. It is inside a tube that runs through your body from one end to the other, and a tube open at both ends is not the same as being in. Nothing you eat counts as being inside you until it crosses the wall of that tube and gets into your blood. Today you will find out exactly where that happens, and it is probably not where you think.

The three questions students will ask

THEY ASK	YOU SAY
“Doesn't the stomach do the digesting?”	It does some, mostly on protein, and it kills bacteria with acid. But almost nothing is absorbed there. The small intestine does the real work.
“Why is the small intestine so long?”	Because absorption needs surface area. Longer tube, folded and covered in tiny projections, means far more surface for food to cross.
“If fiber isn't digested, why eat it?”	Because it is not digested. It gives the muscles in the tube wall something solid to grip and push against, which keeps everything moving.

FINISHED EARLY?

The Go Further page gives absorption figures for a patient before and after part of the small intestine was removed. Students work out which nutrient was hit hardest and why.

Read: A Tube That Runs Right Through You

- 1 Your digestive system is one continuous tube, roughly thirty feet long, running from your mouth to the other end. Coiled up, it fits inside you. Laid out flat, it would stretch across two classrooms. And here is the strange part: anything inside that tube is technically still outside your body, in the same way that a coin dropped through a pipe is not part of the pipe.
- 2 Nothing counts as being *in* you until it crosses the wall of the tube and enters your blood. That crossing is the whole point of digestion, and it explains why food has to be broken down at all. The molecules in a slice of bread are far too large to pass through a cell membrane. Digestion is a size problem before it is anything else.
- 3 There are two ways of making food smaller and both are needed. **Mechanical** breakdown is physical: teeth cutting and grinding, the stomach churning. It does not change what the food is, only how big the pieces are. **Chemical** breakdown does change it, cutting large molecules into smaller ones.
- 4 The chemical work is done by **enzymes**, which are substances that speed up one specific reaction and ignore everything else. There is an enzyme for starch, a different one for protein, another for fat. This is why chewing matters more than it seems: smaller pieces have far more surface exposed, and enzymes can only work on a surface they can reach.
- 5 The stomach is a muscular bag that adds acid, churns for a few hours, kills most of the bacteria that arrived with the meal and begins breaking down protein. What it does *not* do is absorb. Almost nothing crosses into your blood from the stomach, which surprises nearly everyone.
- 6 The absorption happens next door, in the **small intestine**, and this is where the design gets impressive. It is about twenty feet long, its inner surface is thrown into deep folds, and every fold is carpeted with millions of tiny finger-shaped projections called villi. Flattened out, the absorbing surface comes to roughly three hundred square feet, about the floor of a small bedroom, packed into your abdomen.
- 7 What reaches the **large intestine** is mostly water and material that could not be digested. Water is reclaimed here rather than wasted, trillions of harmless bacteria live in it, and what is left is compacted for disposal. Fiber never gets digested at all, and that is exactly why it is useful: it gives the muscles in the tube wall something firm to grip and squeeze against.
- 8 Two organs help without ever having food pass through them. The liver produces bile, which breaks fat into droplets small enough for enzymes to attack. The pancreas sends in a mixture of enzymes for starch, protein and fat. Both deliver chemicals into the tube from the outside, like a factory feeding a production line it does not sit on.

Questions: The Long Journey

Answer in the space provided.

1 Complete the table. Put the organs in the order food meets them, then give the main job of each.

ORDER	ORGAN	WHAT IT MAINLY DOES
	Small intestine	
	Mouth	
	Large intestine	
	Stomach	
	Esophagus	

2 Explain what is meant by saying that food in your stomach is still outside your body.

3 Give two examples of mechanical breakdown and two examples of chemical breakdown.

4 Explain why chewing your food properly makes enzymes work faster. Use the word *surface*.

5 Where is almost all food absorbed into the blood, and name one feature of that organ that makes it possible.

6 Fiber passes through you without being digested at all. Explain why it is still worth eating.

Go Further: The Shortened Tube

THE SITUATION

A patient has a section of small intestine removed after an injury. Doctors measure how much of each nutrient is absorbed from a standard meal, before the operation and six weeks after it. A higher percentage means more of the nutrient got into the blood.

NUTRIENT	ABSORBED BEFORE SURGERY	ABSORBED AFTER SURGERY	CHANGE
Fat	95%	62%	
Protein	92%	78%	
Vitamin B12	88%	30%	
Water	90%	84%	

- 1 Complete the change column. Which nutrient was affected most, and which least?
- 2 Explain why removing a length of small intestine would reduce absorption at all. Use the idea of surface area.
- 3 Vitamin B12 is absorbed only in one particular section near the end of the small intestine. Use that fact to explain why its figure fell so much further than the others.
- 4 Water absorption barely changed. Suggest why, using what you know about the large intestine.

CHALLENGE

The surgeon tells the patient that over the next year the remaining intestine will adapt: the villi will grow longer and the folds deeper, and absorption will improve without any new length being added. Explain how that could work. Two or three sentences.

Answer Key: The Long Journey

QUESTION PAGE

- 1 Mouth — chewing and starch enzymes begin. 2 Esophagus — carries food to the stomach, no digestion. 3 Stomach — acid, churning, protein begins, kills bacteria. 4 Small intestine — most chemical digestion and almost all absorption. 5 Large intestine — reclaims water and compacts waste.
2. The digestive system is a tube open at both ends running through the body. Food inside it has not yet crossed into any tissue or into the blood, so it is not part of you yet.
This is the idea that makes the rest of the lesson make sense. Full credit for anything built on “it has not crossed the wall yet”.
3. Mechanical: chewing, stomach churning. Chemical: enzymes cutting starch into sugar, bile breaking fat into droplets, stomach acid working on protein.
4. Chewing makes the pieces smaller, which exposes far more surface. Enzymes can only work on surfaces they can reach, so more surface means faster digestion.
5. The small intestine. Any one of: it is very long; its lining is deeply folded; it is covered in villi. All three answers are really the same answer — surface area.
6. It is not digested, so it stays solid, and that gives the muscles in the tube wall something to grip and push against. It keeps everything moving.

GO FURTHER PAGE

NUTRIENT	BEFORE	AFTER	CHANGE
Fat	95%	62%	-33
Protein	92%	78%	-14
Vitamin B12	88%	30%	-58
Water	90%	84%	-6

1. Vitamin B12 was affected most, falling 58 points. Water was affected least, falling 6.
2. Absorption depends on surface area. Removing a length of intestine removes some of that surface, so there is less area for nutrients to cross into the blood.
3. Because B12 has only one place it can be absorbed. Removing that section does not just reduce the surface available, it removes the only site that does the job, so most of the B12 has nowhere to cross even though plenty of intestine remains.
This is the best question on the page. It separates “less surface” from “the specific part that mattered”, and only the second one explains the size of the drop.
4. Because most water is reclaimed in the large intestine, which was not operated on. The part doing that job is still intact.

Challenge. Absorption depends on surface area, and length is only one way of getting it. Longer villi and deeper folds pack more surface into the same length of tube, so the remaining intestine can absorb more without getting any longer.

Full credit for any answer that separates surface area from length. That distinction is the whole lesson.

MARKING SHORTCUT

Mark question 2 on the question page and Go Further question 3. The first shows whether the student understood what digestion is actually for; the second shows whether they can reason about a system rather than recite it.

Very Small Enemies

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students separate bacteria from viruses, follow what the body does when something gets past its barriers, and then read antibody data from a first and second exposure. The lesson targets the single most consequential everyday science mistake there is: asking for antibiotics for a cold.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page.
14-28	Questions 1 to 3. The comparison table is the core.
28-40	Questions 4 to 6.
40-53	Go Further: the antibody data.
53-55	Collect all pages.

SAY THIS TO START

There are more bacteria living on and inside you right now than you have cells of your own. You are outnumbered in your own body, and almost none of them are trying to hurt you. A few are, and your body deals with them so efficiently that you never find out. Today you will work out how it does that, and why the same trick works better the second time.

The three questions students will ask

THEY ASK	YOU SAY
"Are all germs bad?"	No. The overwhelming majority are harmless or actively useful. The bacteria in your gut help digest food and take up space that harmful ones would otherwise use.
"Why won't the doctor give antibiotics for a cold?"	Because a cold is a virus and antibiotics only work on bacteria. Taking them would do nothing for the cold and would help resistant bacteria spread.
"Are viruses alive?"	Scientists genuinely disagree. A virus cannot reproduce, feed or do anything at all on its own. It only does anything once it is inside one of your cells.

FINISHED EARLY?

The Go Further page has antibody measurements from a first and a second exposure to the same disease. Students compare speed and size of the two responses and explain the difference.

Read: Outnumbered, and Mostly Fine

1 Microbes are living things too small to see without a microscope, and they are everywhere: in soil, in water, in the air, on this page, and in enormous numbers inside you. The great majority are harmless. A large number are useful. Only a small fraction cause disease, and those are called **pathogens**.

2 **Bacteria** are single cells. They feed, they grow, and they reproduce by splitting in two, which some can do every twenty minutes. One bacterium becomes a million in under seven hours if nothing stops it. Most are harmless; a minority cause things like strep throat, tetanus and tuberculosis.

3 **Viruses** are not cells and are hundreds of times smaller. A virus is essentially a set of instructions in a protein case, and it can do nothing at all on its own — it cannot feed, grow, move or reproduce. It works by getting inside one of your cells and using that cell's machinery to build copies of itself, usually destroying the cell in the process.

4 That difference has a consequence people run into constantly. **Antibiotics** work by attacking structures that bacteria have and your cells do not, such as the bacterial cell wall. A virus has none of those structures, so antibiotics do nothing to it whatsoever. Taking antibiotics for a cold or flu is not merely useless; it also gives resistant bacteria elsewhere in your body a chance to thrive.

5 Before any of that matters, a pathogen has to get in, and your body is genuinely difficult to enter. Skin is a sealed physical barrier. Mucus in your nose and airways traps particles and moves them back out. Tears wash the eyes and contain a substance that attacks bacteria. Stomach acid destroys most of what you swallow. The usual ways in are cuts, the eyes, and being breathed in past the mucus.

6 If something does get through, **white blood cells** take over. Some of them simply engulf and destroy invaders on sight. Others produce **antibodies**, which are proteins shaped to lock onto one specific pathogen and nothing else. An antibody that fits chickenpox is useless against measles. This is why you can have several different infections at once and why each has to be dealt with separately.

7 Making the right antibody from scratch takes time, usually a week or more, and that week is the week you feel ill. But once the infection is beaten, some of those cells remain as **memory cells**. Meet the same pathogen again and they recognize it immediately and produce the correct antibody in enormous quantity within a day or two — fast enough that you often never develop symptoms at all. That is what immunity means.

8 A **vaccine** is a way of getting the memory cells without the illness. It shows the immune system something that looks like the pathogen but cannot cause the disease — a weakened version, a dead one, or just one distinctive piece of it. The body responds as if to a real infection, makes antibodies, and keeps the memory cells. The first real encounter then behaves like a second encounter.

Questions: Very Small Enemies

Answer in the space provided.

- 1 Complete the comparison table.

	BACTERIA	VIRUSES
Is it a cell?		
Can it reproduce on its own?		
Do antibiotics work on it?		
One disease it causes		

- 2 A student with a heavy cold asks the doctor for antibiotics and is refused. Explain both why the antibiotics would not help and why taking them anyway would be a bad idea.

- 3 Name three barriers that stop pathogens getting into the body in the first place, and say what each one does.

- 4 Antibodies are described as fitting one specific pathogen. Explain why that matters when you catch a disease you have never had before.

- 5 Explain why the second time you meet a disease you often do not become ill at all.

- 6 A vaccine gives you protection without giving you the disease. Explain how.

Go Further: The Second Time

THE DATA

A person is exposed to the same pathogen twice, several years apart. The amount of the matching antibody in their blood is measured on the days shown. Units are arbitrary — only the comparison matters.

DAYS AFTER EXPOSURE	0	5	10	15	20	30
First exposure	0	2	15	30	22	10
Second exposure	0	40	180	210	190	150

- Which response was **faster**? Quote two numbers from the table that show it.

- Which response was **bigger**? By roughly how many times, at its peak?

- The person felt ill for about a week after the first exposure and felt nothing at all after the second. Explain why, using the data.

- Explain what the memory cells were doing between the two exposures, and why nothing shows on the table for those years.

CHALLENGE

Some vaccines are given as two or three doses weeks apart rather than one. Using the shape of the two rows in the table, explain what the second dose is for and why it is not simply a repeat of the first. Two or three sentences.

IF YOU STILL HAVE TIME

Antibodies are specific to one pathogen. Suggest why a vaccine for one disease gives you no protection at all against a different one, and why flu vaccines in particular have to be reformulated most years.

Answer Key: Very Small Enemies

QUESTION PAGE

1. Bacteria: yes it is a cell; yes it reproduces on its own; yes antibiotics work; strep throat, tetanus or tuberculosis. Viruses: no, not a cell; no, only inside a host cell; no, antibiotics do nothing; common cold, influenza or chickenpox.
2. A cold is caused by a virus, and antibiotics attack structures only bacteria have, so there is nothing for them to work on. Taking them anyway kills off harmless bacteria and leaves resistant ones behind, which makes those harder to treat later.
Both halves are needed for full credit. Most students get the first and stop.
3. Any three: skin as a sealed physical barrier; mucus in the nose and airways trapping particles and carrying them out; tears washing the eyes; stomach acid destroying what is swallowed.
4. Because the body has no matching antibody ready and has to build the right one from scratch, which takes a week or more. That week is the week you feel ill.
5. Memory cells left over from the first infection recognize the pathogen immediately and produce large amounts of the correct antibody within a day or two, so the infection is stopped before it can build up enough to cause symptoms.
6. It shows the immune system something that looks like the pathogen but cannot cause the disease — a weakened or dead version, or one piece of it. The body makes antibodies and keeps memory cells, so a real infection later is treated like a second encounter.

60 FURTHER PAGE

1. The second. By day 5 the second response had reached 40 while the first was only at 2. Any sensible pair of numbers is fine.
2. The second, by about seven times. It peaked at 210 against 30 for the first.
3. First time, the antibody level stayed very low for over a week, so the pathogen had time to multiply and cause symptoms. Second time, the level was already at 180 by day 10, high enough to stop the infection before it could build up.
Look for use of actual numbers. A student who explains it correctly without quoting the table has given a good answer to a different question — three quarters credit.
4. The memory cells were simply present and inactive, waiting. Nothing shows on the table because the table measures antibody, and memory cells do not produce antibody until the pathogen reappears.

Challenge. The first dose behaves like the first row: slow and small, but it leaves memory cells. The second dose then behaves like the second row, producing a much faster and far larger response. The second dose is not a repeat — it is deliberately triggering the stronger reaction that only becomes possible once memory cells exist.

This is the payoff question. Full credit needs the idea that the second dose gets a different kind of response, not more of the same one.

IF A STUDENT FINISHES EVERYTHING

Ask them why a person can catch a cold many times in their life even though they clearly make memory cells each time. The answer — that “the common cold” is over two hundred different viruses — usually causes a satisfying pause.

The Same Bones

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. Keep: this page and the answer key.

What this lesson does

Students compare the bones inside four very different limbs, find the same arrangement in all of them, and work out what that implies. The lesson is about *evidence* rather than mechanism, and closes on a point that generalises far beyond biology: independent methods agreeing is what makes a conclusion strong.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page and study the limb diagram.
14-28	Questions 1 to 3.
28-40	Questions 4 to 6.
40-53	Go Further: the bone-count table and the mystery fossil.
53-55	Collect all pages.

SAY THIS TO START

Hold your arm out. One bone from your shoulder to your elbow, two bones from your elbow to your wrist, a cluster of small bones in the wrist, then five fingers. Now here is the odd part. A bat's wing has exactly that. So does a whale's flipper. So does a cat's front leg. Four completely different jobs — grabbing, flying, swimming, walking — and underneath, the same bones in the same order. Today you work out why.

The three questions students will ask

THEY ASK	YOU SAY
"Couldn't that just be the best design?"	It is a fair question and worth taking seriously. But a flipper gains nothing from having five separate finger bones sealed inside it. It looks far more like something inherited than something chosen.
"Is this the same as natural selection?"	No. Natural selection is <i>how</i> populations change. This lesson is about the evidence that they are related at all. Different question.
"How do we know the fossils are in the right order?"	Because the order is the same everywhere in the world, on every continent, in rocks studied by people who were not working together.

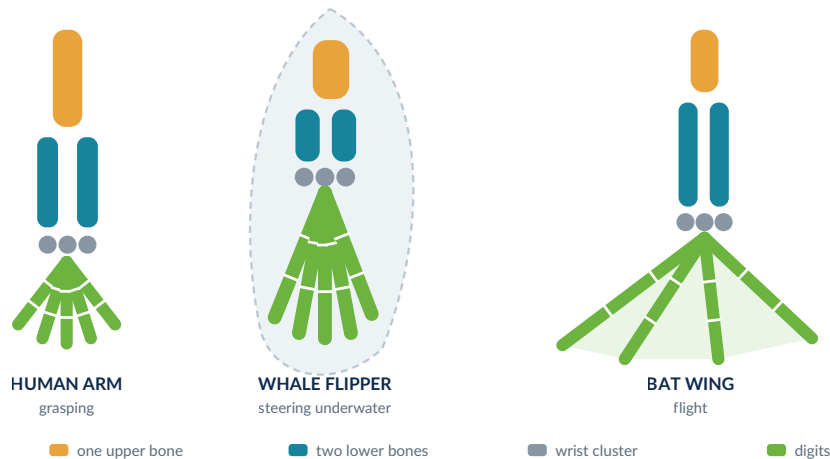
FINISHED EARLY?

The Go Further page has a bone-count table for five animals and a mystery fossil to place using it.

Read: Four Limbs, One Blueprint

BEFORE YOU READ

Three limbs that do three completely different jobs. The colors show which bone is which. Look at the diagram before you read a word of the passage.



1 A human arm, a bat's wing and a whale's flipper look nothing alike from the outside and do nothing alike. Inside, they are the same. One long bone at the top. Two bones below it. A cluster of small wrist bones. Then digits. Change the proportions and you change the job, but the arrangement never changes.

2 Structures like these are called **homologous**: the same underlying structure used for different purposes. And they raise an awkward question. If each limb had been designed from scratch for its job, why would a flipper contain five separate finger bones? Sealed inside a stiff paddle, they do nothing a single solid plate would not do better.

3 The straightforward explanation is that none of these were designed from scratch. All of them are modified versions of the same inherited limb, passed down from a shared ancestor and then stretched, shortened and reshaped by each group over a very long time.

4 Compare that with a **analogous** pair: a bird's wing and a bee's wing. Both fly, and neither has anything in common inside. The bird's wing has bones; the bee's has none at all. Same job, completely different construction, which is what you would expect from two solutions arrived at separately. Doing the same job is not evidence of relatedness. Having the same parts is.

5 A second line of evidence comes from **embryos**. Very early in development, vertebrates that end up wildly different look strikingly similar, and some share structures that later vanish or get repurposed. Human embryos develop a tail, which is reabsorbed before birth. The developing plan is shared even when the finished animal is not.

6 A third line comes from parts that are still present but no longer used, called **vestigial** structures. Whales have small hip bones buried in their bodies, attached to no leg. Some flightless birds have full wing bones. Humans have a tailbone. These are difficult to explain as deliberate features and easy to explain as leftovers.

7 The fourth line is the fossil record. Fossils do not appear in a random order. Simple sea organisms appear first, then fish, then amphibians, then reptiles, then mammals and birds, and the order is the same on every continent. Nobody has ever found a rabbit in rock older than the first mammals, and given how many millions of fossils have been dug up, that is a striking absence.

8 Here is what actually makes this convincing, and it is worth more than any single fact above. Four separate lines of evidence — bones, embryos, leftovers and rocks — were investigated by different people, in different countries, using completely different methods. They agree. When independent methods that could easily have contradicted each other produce the same answer, that agreement is the strongest thing science ever gets.

Questions: The Same Bones

Answer in the space provided.

1 Complete the table using the diagram on the reading page.

LIMB	WHAT IT IS USED FOR	BONES FROM TOP TO BOTTOM
Human arm		
Whale flipper		
Bat wing		

2 Explain what *homologous* means, in your own words.

3 A bird's wing and a bee's wing both do the same job. Explain why scientists do **not** treat that as evidence that birds and bees are closely related.

4 Whales have small hip bones inside their bodies that are attached to no legs at all. Explain what that suggests about whale ancestors.

5 Nobody has ever found a rabbit fossil in rock older than the first mammals, on any continent. Explain why scientists find that absence important.

6 Four different lines of evidence all point the same way. Explain why that is stronger than having one very good line of evidence.

Go Further: Counting the Bones

THE DATA

Bone counts for the front limb of five living animals. Every one of these was measured independently, and none of these animals lives anything like the same life.

ANIMAL	UPPER LIMB BONES	LOWER LIMB BONES	WRIST BONES	DIGITS
Human	1	2	8	5
Whale	1	2	6	5
Bat	1	2	7	5
Cat	1	2	7	5
Frog	1	2	6	4

- 1 Two columns are identical for every animal in the table. Which two, and what does that suggest?
-
-
-
- 2 The wrist and digit columns vary a little. Does that weaken the pattern? Explain.
-
-
-
- 3 A fossil limb is found with **1** upper bone, **2** lower bones, **7** wrist bones and **5** long thin digits spread out like a fan. What can you reasonably conclude, and what can you not conclude?
-
-
-
-
- 4 A dragonfly wing is examined and found to contain no bones at all. Explain what that tells you about how dragonflies are related to the five animals in the table.
-
-
-

CHALLENGE

Someone argues that the shared arrangement is not evidence of a common ancestor at all — it is simply the best possible design for a limb, so anything designing limbs would arrive at it. Give one observation from this lesson that argues against that, and explain why it counts. Two or three sentences.

Answer Key: The Same Bones

QUESTION PAGE

1. Human arm: grasping and manipulating. Whale flipper: steering and swimming. Bat wing: flight. All three: one upper bone, then two bones, then a wrist cluster, then digits.

2. Homologous means the same underlying structure appearing in different species, used for different jobs.

3. Because they are built completely differently inside. A bird's wing has bones and a bee's has none. Doing the same job only shows that both solved the same problem; having the same parts is what suggests shared ancestry.

The distinction between homologous and analogous is the whole point of the question. Accept any answer that gets to "same job is not the same as same structure".

4. That whale ancestors had legs. Hip bones with no legs attached are very hard to explain as a designed feature and easy to explain as a leftover from an animal that walked.

5. Because with millions of fossils collected worldwide, a single one in the wrong layer would be easy to find if the order were random. The order holding everywhere is a prediction that could have failed and has not.

Strong students say something like "it could have been proved wrong and wasn't". That is exactly right and worth extra credit.

6. Because the four methods are independent — different people, different tools, different kinds of evidence. Any one of them could have contradicted the others and none did. Agreement between methods that had no reason to agree is much harder to explain away than one strong result.

GO FURTHER PAGE

1. The upper limb column (always 1) and the lower limb column (always 2). It suggests all five inherited the same basic limb arrangement rather than each developing one separately.

2. No. Small variation is exactly what you would expect if a shared plan has been modified for different jobs over a long time. An identical count in every species would actually be harder to explain.

This question is worth marking. Students often assume any variation ruins a pattern. The point is that a modified inheritance predicts variation around a fixed core.

3. You can reasonably conclude it belonged to an animal in the same broad group as the others, because it has the same arrangement. You cannot conclude which animal it was, or that it definitely flew — long spread digits are suggestive but a swimming or gliding animal could look similar.

4. That the dragonfly is not closely related to any of them, at least not in the way they are related to each other. Its wing does the same job as a bat's but is built on a completely different plan.

Challenge. Best answer: a flipper gains nothing from five separate finger bones sealed inside it, and a whale gains nothing from hip bones attached to no legs. A design chosen fresh for the job would not include parts that do nothing. Leftovers are what inheritance predicts and what design does not.

Accept the whale hips, the flipper digits, the human tailbone or the flightless bird's wing bones. Any answer that identifies a part which is present but useless gets full credit.

MARKING SHORTCUT

Mark question 3 and question 6. Question 3 shows whether the student can tell homologous from analogous, which is the biology. Question 6 shows whether they understood why independent agreement matters, which is the thinking skill and is worth more.

Why Nothing Falls Down

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students take apart the belief that there is no gravity in space, then build the correct picture of an orbit as continuous falling. The lesson finishes with real planetary data showing that everything further from the Sun moves slower, which is gravity weakening with distance made visible.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages.
6-14	Students read the page and study the cannon diagram.
14-28	Questions 1 to 4.
28-40	Questions 5 and 6.
40-53	Go Further: the planetary data table.
53-55	Collect all pages.

SAY THIS TO START

Astronauts on the space station float around, so obviously there is no gravity up there. Except that the station orbits about 250 miles up, and at that height Earth's gravity is still about ninety percent as strong as it is in this room. So gravity is not the explanation. Today you will work out what actually makes them float, and the answer is that they are falling. All of them. Constantly. And missing.

The three questions students will ask

THEY ASK	YOU SAY
"Is there gravity in space?"	Yes, and lots of it. At the space station's height it is about 90 percent of what you feel standing here. Gravity gets weaker with distance but never switches off.
"Then why do astronauts float?"	Because they are falling, and so is the station, and so is everything inside it, all at the same rate. Nothing presses against anything, and that feels exactly like weightlessness.
"Why doesn't the Moon fall into us?"	It is falling. It is also moving sideways fast enough that Earth curves away underneath it as fast as it drops. That is all an orbit is.

FINISHED EARLY?

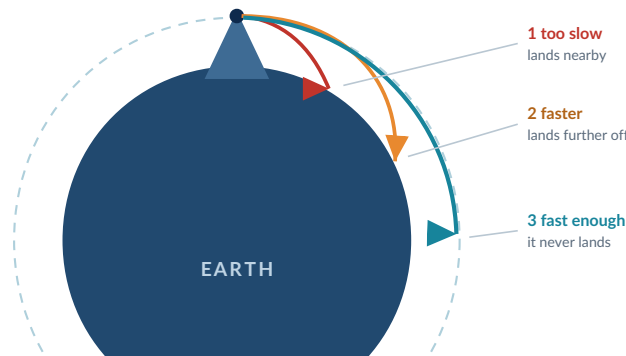
The Go Further page has distance, year length and orbital speed for five planets. Students describe both patterns and then explain the second one using gravity.

Read: Falling, and Missing

BEFORE YOU READ

This is Isaac Newton's thought experiment, drawn in his own book in 1687. A cannon sits on an impossibly tall mountain and fires sideways, harder each time. Question 3 depends on it.

FIRE IT SIDEWAYS, HARDER EACH TIME



All three balls are falling the whole way. Only number 3 is moving sideways fast enough that the ground curves away as fast as it drops.

1 Gravity is a pull between any two objects that have mass. Not just planets — you and this book are pulling on each other right now, far too weakly to notice. It is always a pull, never a push, it cannot be switched off or blocked, and it reaches out forever, getting weaker with distance but never quite reaching zero.

2 Two things decide how strong it is. More mass means a stronger pull, which is why the Earth's grip on you is obvious and this book's is not. More distance means a weaker pull, and it falls away quickly: double the distance and the pull drops to a quarter.

3 Now the space station. It orbits about 250 miles up. Earth is nearly 4,000 miles in radius, so the station has barely moved away from us in proportional terms, and gravity there is still roughly ninety percent as strong as it is in this room. Whatever is making the astronauts float, it is not an absence of gravity.

4 What is actually happening is that they are falling. The station is falling toward Earth. The astronaut is falling. The pen, the water and the sandwich are all falling, at the same rate, at the same moment. Nothing is pressing on anything else, because pressing requires one thing to be held up while another rests on it. That sensation of weight you feel right now is the floor stopping you falling. Remove the floor and the feeling goes, gravity or no gravity.

5 So why does the station not hit the ground? Look at the diagram. Fire the cannon gently and the ball curves down and lands nearby. Fire it harder and it lands further away, because it is still falling at the same rate but traveling further sideways while it does. Fire it hard enough and something remarkable happens: the ground curves away beneath it exactly as fast as it falls, and it never lands at all.

6 That is an **orbit**. Not a place where gravity stops, and not a balance between two forces. It is falling with enough sideways speed to keep missing. The space station is doing about 17,500 miles per hour, which is precisely the speed at which Earth curves away from you as fast as you drop toward it.

7 The Moon is doing the same thing on a larger scale, and has been for four and a half billion years. It falls toward Earth continuously and moves sideways continuously, and the result is a circle. Newton's genuinely radical idea was that the force pulling an apple off a tree and the force holding the Moon are not two forces that happen to be similar. They are one force.

8 Scale that up and you get the solar system. The Sun holds about 99.8 percent of all the mass there is in it, so everything else orbits the Sun rather than the other way round. And because gravity weakens with distance, the planets further out are pulled more gently, move more slowly, and take very much longer to go round.

Questions: Why Nothing Falls Down

Answer in the space provided.

- 1 A student writes: "Astronauts float because there is no gravity in space." This is wrong twice. Explain both errors.

This is the most important question on the page.

- 2 Name the two things that decide how strong the pull of gravity is between two objects.

- 3 Using the cannon diagram, explain in your own words what makes ball 3 different from balls 1 and 2.

- 4 Explain what the feeling of weight actually is. Why does it disappear when you are falling?

- 5 The Moon has been falling toward Earth for four and a half billion years and has never arrived. Explain why.

- 6 The Sun contains 99.8% of the mass of the solar system. Explain what that has to do with the fact that planets orbit the Sun rather than the Sun orbiting the planets.

Go Further: Five Planets, One Pattern

THE DATA

Real measurements for five planets. Distance is given in millions of miles, year length is how long one orbit takes in Earth days, and speed is how fast the planet moves along its orbit.

PLANET	DISTANCE FROM SUN	LENGTH OF ITS YEAR	ORBITAL SPEED
Mercury	36	88 days	29.7 mi/s
Venus	67	225 days	21.7 mi/s
Earth	93	365 days	18.5 mi/s
Mars	142	687 days	15.0 mi/s
Jupiter	484	4,333 days	8.1 mi/s

- Describe the relationship between a planet's distance from the Sun and the length of its year.

- Describe the relationship between distance and orbital speed.

- Jupiter is about 5 times further out than Earth, but its year is nearly 12 times longer, not 5 times. Explain why using your answer to question 2.

- Explain why planets further from the Sun travel more slowly. Use the word gravity.

CHALLENGE

Imagine the Sun's gravity switched off completely and instantly. Earth would not fall into the Sun and it would not stop. Describe the path Earth would actually take from that moment, and explain why. Two or three sentences.

IF YOU STILL HAVE TIME

The space station has to be given a push by a rocket every few months to raise its orbit. There is no air in space, so suggest what is slowing it down — and what would eventually happen if nobody ever pushed it again.

Answer Key: Why Nothing Falls Down

QUESTION PAGE

1. Error one: there is gravity in space. At the space station's altitude it is about 90 percent as strong as at ground level. Error two: even if it were weaker, that is not why they float. They float because they and the station are falling together, so nothing presses on anything.

Most students get error one and stop. Error two is the deeper one and is the point of the lesson. Half credit if only one is given, and write the other one in.

2. The masses of the two objects, and the distance between them.

3. All three balls fall at exactly the same rate. Balls 1 and 2 are not moving sideways fast enough, so the ground is still underneath them when they arrive. Ball 3 is moving sideways fast enough that the Earth curves away beneath it as fast as it falls, so it never reaches the ground.

Full credit needs "all three are falling equally". A student who thinks ball 3 stops falling has the common misconception the diagram exists to remove.

4. Weight is the sensation of something — the floor, a chair, the ground — pushing up and stopping you falling. When you are falling freely nothing is stopping you, so there is nothing pressing on you and the feeling disappears.

5. Because it is also moving sideways. It falls toward Earth continuously, but its sideways motion carries it far enough that Earth curves away underneath it, so it keeps missing.

6. Gravity is stronger for larger masses. The Sun has so much more mass than everything else combined that its pull dominates, so everything else orbits it.

Strong students note that the Sun does wobble very slightly in response to the planets. That is correct and is how planets around other stars were first detected — worth extra credit.

GO FURTHER PAGE

1. The further from the Sun, the longer the year. Mercury takes 88 days at 36 million miles; Jupiter takes 4,333 days at 484 million.

2. The further out, the slower. Mercury moves at 29.7 miles per second and Jupiter at only 8.1.

3. Because a further-out planet has two separate disadvantages at once. Its orbit is a much longer way round, *and* it is traveling more slowly along it. A longer path at a lower speed multiplies out to far more than five times the time.

This is the best question on the page. Full credit needs both effects. One alone is half credit.

4. Gravity gets weaker with distance, so the Sun pulls a distant planet less strongly. A weaker pull can only bend a slower-moving planet into a circle — a fast planet that far out would not be held and would leave.

Challenge. Earth would travel in a perfectly straight line, in whatever direction it happened to be moving at that instant, and would keep going forever. Nothing would be curving its path any more. An orbit is a straight line being continuously bent by gravity, so removing the gravity leaves the straight line.

The common wrong answers are "it would fall into the Sun" and "it would stop". Both are worth discussing: falling in requires a pull, and stopping requires a push.

How a Mountain Becomes Sand

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students separate three processes that students and adults alike run together, then use a river's speed at four points to predict what it carries and what it drops. The sorting of sediment by size is the payoff, and it makes sense of beaches, deltas and gravel bars in one go.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page.
13-26	Questions 1 to 3. The classification table is the core.
26-38	Questions 4 to 6.
38-48	Go Further: the river at four speeds.
48-50	Collect all pages.

SAY THIS TO START

Pick up a handful of sand at a beach and you are holding a mountain. Not a metaphor — every grain was once part of solid rock somewhere upstream, and it got here by being broken off, carried and dropped. Today you will learn the three separate processes that did that, and why they are three processes and not one.

The three questions students will ask

THEY ASK	YOU SAY
"Isn't weathering just the same as erosion?"	No, and it is the main thing to get right today. Weathering breaks rock where it stands and moves nothing. Erosion is the carrying away. Different jobs, often different times.
"How can water break a rock?"	By freezing. Water expands by about nine percent when it turns to ice, so water that seeps into a crack and freezes pushes the crack wider. Repeat that a thousand winters.
"Why is sand at the beach and gravel upstream?"	Because a river drops the heaviest pieces first as it slows. By the time it reaches the sea it can only still be carrying the smallest ones.

FINISHED EARLY?

The Go Further page gives one river at four points with four different speeds, plus a table of how fast water has to move to carry each size of particle. Students work out what is dropped where.

Read: Broken, Carried, Dropped

- 1 Two opposite things happen to the Earth's surface at once. Mountains are pushed up, slowly and enormously. And everything that is up gets worn back down. This lesson is entirely about the wearing down, which is quieter, much less dramatic, and never stops for a moment.
- 2 It comes in three stages, and running them together is the mistake almost everyone makes. **Weathering** is rock being broken apart *where it sits*. Nothing is transported. **Erosion** is the broken pieces being carried away. **Deposition** is those pieces being put down somewhere else. Weathering makes the pieces, erosion moves them, deposition parks them.
- 3 **Physical weathering** breaks rock without changing what it is made of. The most effective version in cold places is freeze-thaw. Water seeps into a crack, freezes, and expands by about nine percent as it does. That expansion levers the crack a little wider. It melts, seeps deeper, and freezes again. A thousand winters of that will split a boulder, and nothing about the rock has chemically changed.
- 4 **Chemical weathering** changes the rock itself. Rainwater is naturally slightly acidic, and limestone dissolves in it. Given long enough, that is how caves and sinkholes form — a hole eaten through solid rock by very weak acid and a great deal of patience. Iron minerals rust and crumble by the same kind of process.
- 5 Living things do it too. Tree roots grow into cracks and push them apart with real force. Lichens release weak acids onto bare rock. Burrowing animals expose fresh surfaces to the weather. It is slow, but nothing about this topic is fast.
- 6 Once pieces are loose, four things move them: **water**, which does far more than the other three combined; **wind**, which matters most where there is little plant cover; **ice**, where glaciers grind and drag material for miles; and **gravity** alone, in rockfalls and landslides. What they all have in common is energy. More energy means bigger pieces moved.
- 7 Which sets up the whole of deposition. A moving river carries whatever it has enough energy to carry. When it slows — at a bend, on a flat plain, where it meets the sea — it can no longer hold the heaviest material, and that material drops first. Slow further and the next size down drops. This is why sediment ends up **sorted**: boulders and gravel upstream, sand in the middle reaches, and fine silt and clay at the mouth.
- 8 Rates are almost comically small. A mountain range might lose a fraction of an inch a year. But run that for ten million years and you have removed miles of rock. The processes in this lesson are not weak. They are just extremely patient, and they have never once stopped.

Questions: How a Mountain Becomes Sand

Answer in the space provided.

1 Classify each example. Write **W** for weathering, **E** for erosion or **D** for deposition.

#	WHAT HAPPENS	W, E OR D
1	A crack in a cliff widens after a hard frost	
2	A river drops gravel on the inside of a bend	
3	Wind carries desert sand across a valley	
4	Rainwater slowly dissolves a limestone hillside	
5	A glacier drags boulders twenty miles down a valley	
6	Sand builds up at the mouth of a river	
7	A tree root splits a paving stone in half	

2 Explain the difference between weathering and erosion in one sentence.

3 Describe, step by step, how freeze-thaw breaks a rock apart. Include what water does when it freezes.

4 Rainwater is only very weakly acidic. Explain how something that weak can hollow out a cave.

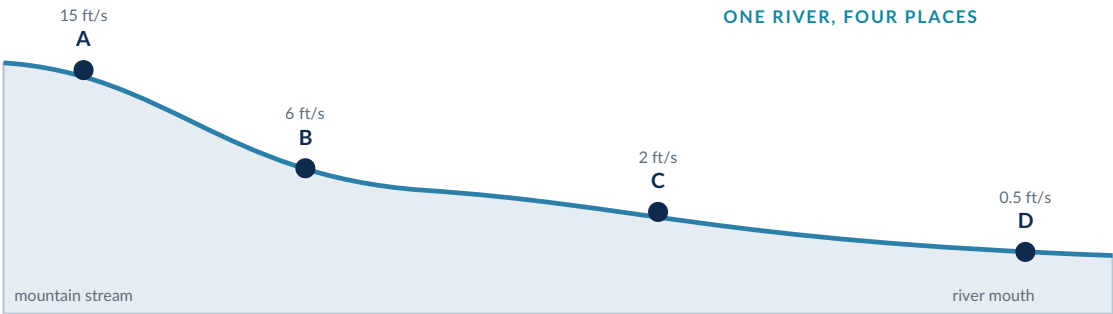
5 Explain why a fast-flowing river can carry bigger pieces than a slow one.

6 A mountain range wears down by less than a hundredth of an inch a year. Explain why that is still enough to matter.

Go Further: What the River Drops, and Where

THE SITUATION

One river, followed from a mountain stream to the sea. The speed of the water is given at four points.



How fast the water has to move to keep each particle moving:

PARTICLE	BOULDER	PEBBLE	SAND	SILT	CLAY
Speed needed	13 ft/s	3 ft/s	1 ft/s	0.3 ft/s	0.1 ft/s

1 Complete the table. At each point, list what the river can still carry and what it must drop.

POINT	SPEED	STILL BEING CARRIED	DROPPED HERE
A	15 ft/s		
B	6 ft/s		
C	2 ft/s		
D	0.5 ft/s		

2 Explain why a beach at the river mouth is made of sand and silt rather than boulders.

3 After a storm the river at point C speeds up to 8 ft/s. What changes, and what happens to material that had already been deposited there?

CHALLENGE

A dam is built across the river between point B and point C. Water still flows out of the dam, but very slowly inside the reservoir behind it. Predict what happens to the sediment behind the dam, and what happens to the beach at the river mouth over the next fifty years. Explain both. Three or four sentences.

Answer Key: How a Mountain Becomes Sand

QUESTION PAGE

1. 1 W, 2 D, 3 E, 4 W, 5 E, 6 D, 7 W.

Item 5 is the one to watch. A glacier both weathers and erodes, but the sentence describes dragging, which is transport, so E is the answer. If several students write W, that is a good sign they are thinking rather than guessing.

2. Weathering breaks rock apart where it is; erosion carries the broken pieces away.

3. Water seeps into a crack. It freezes and expands by about nine percent, forcing the crack wider. It melts and seeps deeper into the now-larger crack. It freezes again. Repeated many times, the rock splits.

Full credit needs the expansion on freezing and the repetition. One freeze does almost nothing.

4. By working on the same spot for an extremely long time. A very weak acid removes a very small amount each year, but over hundreds of thousands of years that adds up to a cave.

5. Because it has more energy. Moving a bigger piece takes more energy, so only faster water can keep the largest material moving.

6. Because it never stops and the timescales are enormous. Less than a hundredth of an inch a year for ten million years still removes over a mile of rock.

60 FURTHER PAGE

POINT	SPEED	STILL CARRIED	DROPPED
A	15 ft/s	Everything, including boulders	Nothing
B	6 ft/s	Pebbles, sand, silt, clay	Boulders
C	2 ft/s	Sand, silt, clay	Pebbles
D	0.5 ft/s	Silt and clay	Sand

2. Because boulders were dropped far upstream, as soon as the river slowed below 13 ft/s. By the mouth the water is moving at 0.5 ft/s and can only still be carrying the very smallest particles, so only those arrive.

3. At 8 ft/s the river can carry pebbles again, and it will pick up pebbles it had previously deposited there. It still cannot move boulders, which need 13 ft/s. So a storm both carries more and re-erodes material that had already been dropped.

The second half is the interesting part. Deposition is not permanent – it lasts only as long as the river stays slow.

Challenge. Behind the dam the water is almost still, so it can carry nothing at all and drops everything. Sediment builds up in the reservoir, which slowly fills with mud and gravel. Downstream, the water leaving the dam is carrying almost no sediment, so nothing new reaches the mouth. The beach keeps being eroded by waves but is no longer resupplied, so it shrinks and may disappear.

This is a genuinely difficult question and a real one – it has happened to beaches below major dams worldwide. Full credit needs both halves: build-up behind, starvation below. Half credit for either alone.

MARKING SHORTCUT

Mark question 2 and the Challenge. Question 2 tests the one distinction the whole lesson exists for. The Challenge tests whether they can run the sorting idea forward into a situation nobody described to them.

Emergency Day Checklist

PRINT THIS PAGE AND LEAVE IT ON TOP

Fill in the blanks once, at the start of the year, and this page works for every absence after that.

Note for the substitute

My name

Room / period

Teach this lesson Lesson #_____ or any lesson from the index

Teacher next door

Lab / storeroom policy Keep it closed. Nothing in this pack needs equipment.

Students who leave early

Accommodations Seating, extra time, reader

Fire drill meeting point

Leave the work

Anything else

SUBSTITUTE: BEFORE YOU LEAVE, DO THESE THREE THINGS

1. Stack the collected work on the teacher's desk with this page on top.
2. Write the names of two or three students who were especially helpful. Teachers genuinely act on this.
3. Note anything the teacher should know, in one line. Not a report. One line.

Substitute's note back to the teacher

Lesson taught

Got through All of it / Most of it / Page 1 only

Helpful students

Anything to know

REUSE NOTE

This pack is designed to be printed once and reused across many absences. Store the briefing pages and answer keys separately from the student pages so the wrong sheets never get photocopied.

Standards Alignment

NGSS MIDDLE SCHOOL · GRADES 6-8

Each lesson is mapped to the closest matching middle school performance expectation. Most lessons also touch others; the primary alignment is listed for clarity.

#	LESSON	FOCUS	NGSS
21	Bending the Beam	Light: reflection, refraction and how we see	MS-PS4-2
22	Fields You Cannot See	Magnets, magnetic fields and electromagnets	MS-PS2-3
23	The Scale That Sorts	Acids, bases, the pH scale and neutralization	MS-PS1-2
24	Made, Not Found	Synthetic materials, polymers and trade-offs	MS-PS1-3
25	The Gap Before You Move	Nervous system, stimulus and response, reflexes	MS-LS1-8
26	The Long Journey	Digestive system, enzymes and absorption	MS-LS1-3
27	Very Small Enemies	Microbes, disease, immunity and vaccines	MS-LS1-1
28	The Same Bones	Evidence for common ancestry	MS-LS4-3
29	Why Nothing Falls Down	Gravity, orbits and the solar system	MS-ESS1-2
30	How a Mountain Becomes Sand	Weathering, erosion and deposition	MS-ESS2-2

Coverage by domain

DOMAIN	LESSONS	WHAT IS COVERED
Physical Science	21, 22, 23, 24	Reflection and refraction, magnetic fields and electromagnets, the pH scale and neutralization, synthetic materials and polymers
Life Science	25, 26, 27, 28	The reflex arc and reaction time, digestion and absorption, microbes and immunity, evidence for common ancestry
Earth and Space Science	29, 30	Gravity, orbits and the solar system; weathering, erosion and deposition

HOW THIS PACK RELATES TO THE REST OF THE SERIES

Volumes 1 and 2 cover thirty different lessons between them, including forces, energy, the particle model, the periodic table, cells, food webs, waves, heredity, natural selection, circuits, density, weather, plate tectonics and rock layers. Nothing in Volume 3 repeats any of them. Together the three volumes give thirty emergency science lessons with no overlap.

This resource is independently created and is not affiliated with, endorsed by, or reviewed by any state agency or standards body. NGSS codes are provided to help teachers document the day's instruction.

Terms of Use and Credits

THANK YOU

Thank you for buying Volume 3. It was written by a practicing teacher with more than twenty years in secondary classrooms, and every lesson is built the way I would want a sub plan built if I were the one waking up sick.

If it saved you a morning, a rating on the listing genuinely helps other teachers find it, and earns you credit toward your next purchase.

Getting the most out of this pack

Print it before you need it.	The whole point is that it works on a morning when you cannot get out of bed to print anything.
Store it in two piles.	Student pages in one, briefing pages and answer keys in the other, so the wrong sheets never get copied by mistake.
Tell your office where it is.	On a genuine emergency day, the person who needs to find this is not you.
Use them when you are present too.	These work as standalone skill lessons, review days, and non-lab days when the lab is booked out.

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OTHER PACKS IN THIS SERIES

- Emergency Sub Plans · Middle School Science, **Volume 1** · ten different lessons, no lab required, 57 pages
- Emergency Sub Plans · Middle School Science, **Volume 2** · ten more lessons, no lab required, 57 pages
- Emergency Sub Plans · Middle School Math · ten no-prep lessons, 57 pages
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